Physical Models of Biological Systems 77972

Formula and concept sheet

Model formulation and inference

Cycle: formulate → solve → validate → revise.
Uses: consistency, prediction, inference, intuition.
Checks: units, signs, initial values, positivity, limits.
Steady state: constant amounts; fluxes may persist.
Identifiable: distinct parameters give distinct data.
Over-constrained: more independent constraints than free parameters. Does not ensure identifiability.
Overfit: excess flexibility fits noise. Count independent features; correlated points add less information.

Production, clearance, and stability

$$\dot{W} = Q_{\text{in}} - Q_{\text{out}}, \qquad \dot{W} = 0 \Rightarrow Q_{\text{in}} = Q_{\text{out}}$$
$$\dot{W} = Q - \kappa W, \qquad W^* = Q/\kappa$$
$$W(t) = W^* + [W(0) - W^*]\mathrm{e}^{-\kappa t}$$

$[Q] = [W]/\text{time}$, $[\kappa] = \text{time}^{-1}$.
Stable for $\kappa > 0$: deviations decay; $\tau = 1/\kappa$.
Steady amount fixes a ratio, not either flux.

$$\Delta p = \rho g h, \quad W = Ah, \quad Q_{\text{out}} \propto \Delta p \propto W$$

Fixed area A; density ρ; viscous Poiseuille outlet.

First-order decay and semilog plots

$$\dot{N} = -kN \Rightarrow N = N_0\mathrm{e}^{-kt}$$
$$\tau = 1/k, \qquad t_{1/2} = \ln 2/k$$
$$\ln N = \ln N_0 - kt, \qquad \frac{\mathrm{d}\log_{10} N}{\mathrm{d}t} = -\frac{k}{\ln 10}$$
$$r_{\text{app}} = \frac{\ln[N(t_1)/N(t_2)]}{t_2 - t_1}$$

For multiple modes, r_{app} need not be a process rate.

Linear forced equation: master method

$$\dot{y} + a(t)y = f(t), \quad \mu(t) = \mathrm{e}^{\int_0^t a(s)\mathrm{d}s}$$
$$(\mu y)' = \mu f, \quad y(t) = \frac{y(0) + \int_0^t \mu(s) f(s)\mathrm{d}s}{\mu(t)}$$

Two-stage viral turnover

Assume: no new infections after $t = 0$; constant rates; mixed compartment; expected populations.
N_I: infected cells; N_V: total free particles, including noninfectious ones; N_{I0}, N_{V0}: initial values.
k_I, k_V: clearance rates [time^{-1}]; γ: virions/(cell time); $\beta = \gamma N_{I0}$: virions/time. RNA load measures abundance, not infectivity.

$$\dot{N}_I = -k_I N_I, \qquad N_I = N_{I0}\mathrm{e}^{-k_I t}$$
$$\dot{N}_V = \gamma N_I - k_V N_V = \beta \mathrm{e}^{-k_I t} - k_V N_V$$

For $k_V \neq k_I$:

$$N_V = N_{V0}\mathrm{e}^{-k_V t} + \frac{\beta(\mathrm{e}^{-k_I t} - \mathrm{e}^{-k_V t})}{k_V - k_I}$$

Pretreatment balance: $\beta = k_V N_{V0}$.

$$\frac{N_V}{N_{V0}} = \frac{k_V \mathrm{e}^{-k_I t} - k_I \mathrm{e}^{-k_V t}}{k_V - k_I}$$

$$\dot{N}_V(0) = 0, \qquad \ddot{N}_V(0) = -k_I k_V N_{V0}$$
$$N_V/N_{V0} = 1 - \tfrac{1}{2} k_I k_V t^2 + O(t^3)$$

Shoulder: horizontal tangent, not a finite flat interval. For $k_I = k_V = k$:

$$N_V = (N_{V0} + \beta t)\mathrm{e}^{-kt}$$
$$\beta = kN_{V0} \Rightarrow N_V = N_{V0}(1 + kt)\mathrm{e}^{-kt}$$

Equal rates still give zero initial slope.

$$k_V \gg k_I : N_V \simeq (\beta/k_V)\mathrm{e}^{-k_I t}$$
$$k_I \gg k_V : N_V \propto \mathrm{e}^{-k_V t}$$
$$t \to \infty : -\mathrm{d}\ln N_V/\mathrm{d}t \to \min(k_I, k_V)$$

Inference: measured N_{V0} leaves 3 unknowns (β, k_I, k_V); balance reduces to 2.
Symmetry: $k_I \leftrightarrow k_V$ preserves the balanced curve: only the unordered pair is identifiable. Also $\beta = \gamma N_{I0}$ cannot separate γ, N_{I0}.
Scope: early treatment response; omits reservoirs and long-term progression.

Pulse-chase: source and destination

$$\dot{D} = kS - dD, \qquad D = K * S$$
$$D = k\int_{t_0}^{t} \mathrm{e}^{-d(t-t')} S(t')\mathrm{d}t' + \mathrm{e}^{-d(t-t_0)} D_0$$
$$K(\tau) = k\mathrm{e}^{-d\tau}, \qquad \tilde{D}/\tilde{S} = k/(s + d)$$

S, D: label per cell upstream and downstream; k: transfer rate; d: labelled loss rate, both [time^{-1}].
Fit: with $G_d(t_i) = \int_{t_0}^{t_i} \mathrm{e}^{-d(t_i - t')} S\mathrm{d}t'$,

$$k_* = \frac{\sum_i D_i G_i}{\sum_i G_i^2}, \qquad \chi^2_{\min} = \sum_i D_i^2 - \frac{\left(\sum_i D_i G_i\right)^2}{\sum_i G_i^2}$$

Fit the integral, not the slope: differentiating multiplies noise by $\mathrm{i}\omega$, integrating divides by it.
Division dilutes: add $\ln 2\, r_{\text{div}}$ to the effective loss rate.

HIV: infectious and non-infectious virions

$$\dot{N}_U = \lambda - k_U N_U - \epsilon N_V N_U$$
$$\dot{N}_I = \epsilon N_V N_U - k_I N_I$$
$$\dot{N}_V = \epsilon' \gamma N_I - k_V N_V$$
$$\dot{N}_X = (1 - \epsilon')\gamma N_I - k_V N_X$$

ϵ: infection coefficient, [(virion·time)$^{-1}$]; ϵ': infectious fraction, dimensionless.

$$N_U^* = \frac{k_I k_V}{\epsilon \epsilon' \gamma}, \qquad \mathcal{R}_0 = \frac{\lambda \epsilon \epsilon' \gamma}{k_U k_I k_V}$$

Infected state exists only for $\mathcal{R}_0 > 1$. **Protease inhibitor:** $\epsilon' \to 0$ gives $N_V = N_{V0}\mathrm{e}^{-k_V t}$.

Independent resistance barriers

$$P(\text{all } n) = p^n, \quad \mathbb{E}[\text{resistant copies}] = Mp^n$$

p: each mutation probability; M: opportunities. Requires independence and simultaneous mutations; omits sequential evolution and cross-resistance.

Discrete probability

$$\sum_{\ell} P(\ell) = 1, \quad P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

$$P(E \mid E') = \frac{P(E \cap E')}{P(E')}, \quad P(E \cap E') = P(E)P(E') \text{ (independent)}$$

$$P(H \mid E) = \frac{P(E \mid H)P(H)}{\sum_j P(E \mid H_j)P(H_j)}$$

Geometric: $P(J = j) = \xi(1-\xi)^{j-1}$, $\mathbb{E}[J] = 1/\xi$, $\mathrm{Var} = (1-\xi)/\xi^2$.
Binomial: $P(K = k) = \binom{n}{k}\xi^k(1-\xi)^{n-k}$, $\mathbb{E}[K] = n\xi$.
Screening: prevalence π, sensitivity s, specificity c,

$$P(D \mid +) = \frac{\pi s}{\pi s + (1-\pi)(1-c)}$$

Rare disease: a positive test is still mostly wrong.

Entropy and mutual information

$$H(X) = -\sum_x P(x)\log_2 P(x), \qquad 0 \le H \le \log_2 M$$

$$H(Y \mid X) = \sum_x P(x)\, H(Y \mid X = x)$$

$$I(X;Y) = H(X) + H(Y) - H(X,Y) = H(Y) - H(Y \mid X)$$

$$I(X;Y) = \sum_{x,y} P(x,y)\log_2 \frac{P(x,y)}{P(x)P(y)} \ \ge 0$$

$$D_{\mathrm{KL}}(P\|Q) = \sum_x P(x)\log_2 \frac{P(x)}{Q(x)} \ \ge 0$$

H in bits; 2^H is an effective number of states. $I = 0$ iff independent; $D_{\mathrm{KL}} = 0$ iff $P = Q$, and it is asymmetric.
Jensen: $\phi(\mathbb{E}Z) \le \mathbb{E}[\phi(Z)]$ for convex ϕ; taking $\phi = -\log_2$ proves both inequalities.
Bernoulli KL: $D(p\|q) = p\log_2 \frac{p}{q} + (1-p)\log_2 \frac{1-p}{1-q}$.
BSC, error ε: $I = H(Y) - h(\varepsilon) \le 1 - h(\varepsilon)$, $h(\varepsilon) = -\varepsilon\log_2 \varepsilon - (1-\varepsilon)\log_2(1-\varepsilon)$.
Data processing: $X \to Y \to Z$ implies $I(X;Z) \le I(X;Y)$.